(a) Overall architecture

TRAIN captions
Nouns + verbs

F **Lemma → CLIP text**
Frozen noun / verb banks

L **Aggregation ×2**
150 / bank

P_n / P_v
Global semantic prototypes

VISUAL EVIDENCE

I

F **CLIP visual**
Frozen encoder

Patches
14 × 14 × 768
No CLS

ROI pool
N × 768

L **Linear**
768 → 512
V: N × 512

F **YOLO**
Detections

Boxes
+ class labels

CSA

P_n

L **Object Fusion 1**
V, P_n → V_1, T_1

O-links
t_o, i, b_i

L **Token prefix**
Word emb. e + position

V_1, T_1

L **Object Fusion 2**
V_1, T_1 → V_o

V_o

L **Object Adapter**
Panel (b)

V_o'

L **Transformer ×3**
Masked SA → H
Object CA ‖ Relation CA
Q=H; K,V=V_o' / V_r'
$g_o(c_o,e)$, $g_r(c_r,e)$
$g_o \odot c_o + g_r \odot c_r$ → FFN

L **Head → beam 5**
Next-token $p(y_t)$
≤20 tokens
5 captions: $y_1,\ldots,y_5$

V_o

P_v

L **Relation Fusion**
V_o, P_v → V_r

V_r

L **Relation Adapter**
Panel (b)

V_r'

IMAGE-SPECIFIC PROMPTS

F **UPT**
Boxes · object labels
Actions · scores

Filter + construct
Object / relation phrases

F **CLIP text encoder**
t_o, t_r

YOLO boxes

Class + IoU ≥ 0.5
i; ordered pair (s, o)

Linked prompts
Text + ROI + boxes
Valid O-links / R-links

I + YOLO labels

Reranking
D: Σ log p · C: CLIPScore
O: object consistency
S = 0.3D + 0.4C + 0.3O

Caption: argmax S

D, C: min–max over 5 beams · O: normalized · ties: earlier beam

(b) Region-linked prompt adapters

Object adapter

Object slot
$[t_o; V_o[i]; b_i]$
$b_i \in \mathbb{R}^4$

L **MLP**

Slots
K, V

L **Masked MHA**
Q = V_o

L **Wout**
Zero-init

⊕ → V_o'

V_o

$M_o[n,k] = 1 \Leftrightarrow n = i_k$

Relation adapter

Relation slot
$[t_r; V_o[s]; V_o[o]; b_s; b_o]$
$[b_s; b_o] \in \mathbb{R}^8$

L **MLP**

Slots
K, V

L **Masked MHA**
Q = V_r

L **Wout**
Zero-init

⊕ → V_r'

V_r

$M_r[n,k] = 1 \Leftrightarrow n \in \{s_k, o_k\}$

Ordered (s,o) · relation slots use original V_o

d = 512 · 8 heads · Wout = 0 at init · No valid slot ⇒ ΔV = 0

(c) Training objective

Stage 1 · Cross-entropy

Shifted TRAIN caption
Teacher forcing

L **Same caption model**
$p\theta(y^*_t \mid I, y^*_{<t})$

$$L_{CE} = -\text{mean} \log p\theta(y^*_t \mid I, y^*_{<t})$$

Stage 2 · Self-critical training

Same model
Same image I

Sampling → y_s

Greedy → y_g

CIDEr advantage
$A = r(y_s) - r(y_g)$

Ytrain references

$r(y) = \text{CIDEr}(y, Y_{train})$

$$L_{SCST} = -\text{mean_batch}\,[\text{stopgrad}(A) \times \Sigma_t \log p\theta(y_{s,t} \mid I, y_{s,<t})]$$

$L = L_{CE} + 0.05\, L_{SCST}$

Visual · Semantic / prompts · Decoder / training · Reranking

F Frozen L Learned ⊕ Residual sum